Computational Logic — Homework #1

5 October 2026

Due Friday, 16 October 2026. This assignment is *not graded*, but submitting it by the deadline is a condition for sitting the exam. Model answers will be discussed in class.

How to submit. Exercises 2, 3 and 5 go to the Moodle assignment "Homework 1". Exercises 1 and 4 are posted to the Moodle discussion forum instead — posting *is* your submission for those two.

Preferably, typeset your answers in LaTeX. Good luck!

Exercise 1 Find an application of propositional logic in computer science or artificial intelligence and describe it in 150 to 300 words. Post it to the Moodle discussion forum.

Before you post, read what is already there: **choose an application nobody has claimed yet**, and say in one sentence what is specifically *propositional* about it — what the atoms stand for, and what the formulas assert. Then reply substantively to one other post: either an application you had not thought of, or one where you think the logic is doing less work than claimed.

Exercise 2 Every two-element set of connectives containing $\neg$ together with one of $\{\wedge, \vee, \to\}$ can define all of $\{\neg, \wedge, \vee, \to, \leftrightarrow\}$. In each part below you may use connectives you have already defined in an earlier part.

- Define $\vee$, $\to$ and $\leftrightarrow$ using only $\wedge$ and $\neg$.
- Define exclusive disjunction (XOR, $\veebar$) using only $\wedge$ and $\neg$.
- There are in fact two binary connectives, dual to each other, each of which is complete *on its own*. NAND means "not both": $p \uparrow q$ abbreviates $\neg(p \wedge q)$. NOR means "neither nor": $p \downarrow q$ abbreviates $\neg(p \vee q)$. Pick one of the two and show that it suffices to define all of $\{\neg, \wedge, \vee, \to, \leftrightarrow\}$.

Exercise 3 Prove or disprove the validity of the following sequents. At least one of them is *not* valid; for that one, give a valuation and say which theorem from Section 1.4.3 licenses you to conclude from it that no proof exists.

- $p \wedge (q \wedge r) \vdash (p \wedge q) \wedge r$
- $\neg p \wedge \neg q \vdash \neg(p \vee q)$
- $\neg\neg p \vdash p$
- $p \to (q \to r) \vdash (p \to q) \to r$

For the third sequent, note which rule your proof needs, and ask yourself what you could still prove if that rule were taken away. Keep your answer — Exercise 4 is about exactly this.

Exercise 4 What is intuitionistic propositional logic? How does natural deduction for it differ from the system studied in class, and which of the sequents in Exercise 3 would an intuitionist refuse to accept? What are the applications of intuitionistic logic in computer science? Answer in 250 to 500 words and post it to the Moodle discussion forum, following the same rule as Exercise 1: read the thread first, and add something that is not already there.

Exercise 5 Define disjunctive normal form (DNF) formulas. When would it be helpful to have a formula in DNF, and how does that compare with what CNF is good for? Sketch in pseudocode an algorithm that turns any propositional formula into an equivalent DNF formula. Does your algorithm have the same worst-case behaviour as the CNF algorithm from class? Justify your answer briefly.